Matched-Trajectory Evaluation of Photonic-Crystal Tactile Inner-Loop Feedback for Slip-Responsive Grasping

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Abstract

Local contact can redistribute before a teleoperator observes gross slip, while preemptively tightening a grasp increases object loading. Mechanochromic structural-color tactile sensing has demonstrated contact mapping and closed-loop force control, but its slip-responsive robot utility remains unresolved. We study an SO-101 leader–follower system in which a bounded photonic-crystal tactile inner loop overrides only the follower gripper command. A 98,328-parameter causal estimator processes a 100-ms history from two fingertips with a matched $4 \times 32 \times 64$ *input. Six-fold leave-one-material-out cross-fitting locks preprocessing, checkpoint selection, thresholds, and robot-facing models without using the tested surface material. The evaluation comprises 384 frozen-trajectory primary trials, 18 fixed-high references, and 96 trajectory-paired online interventions against matched GelSight, wrist force/torque, and fixed-command baselines. We report [RESULT: event sensitivity, false alarms per minute, and prediction lead], [RESULT: slip-free completion risk difference and restricted-mean time-to-slip difference], and [RESULT: maximum-slip and gripper-command-integral differences]. The study tests robot-level utility within six predefined object–surface pairs; it does not claim a new mechanochromic mechanism, arbitrary-material generalization, or calibrated force without an independent calibration.*

1. Introduction

Low-cost leader–follower manipulators make robot data collection increasingly accessible [1]. Yet local contact remains partly hidden inside the gripper. A scalar command or wrist wrench can remain similar while the contact patch redistributes and an object approaches gross slip. Raising the grip preemptively can improve stability, but it also increases object loading and provides no evidence that the extra action was necessary.

Tactile slip detection and control are established research problems. Commercial-sensor benchmarking has shown strong dependence on material, sampling rate, window size, and sensor variability [2]. Optical and biomimetic fingertips have detected slip and stabilized grasps through local motion or learned features [3, 4]. GelSight-family sensors additionally support shear, incipient-slip, and distributed-force analyses [5, 6, 7, 8]. Event-driven fingertips and recent reflex controllers further demonstrate that fast tactile feedback can regulate gripper action [9, 10]. Thus, closed-loop slip control itself is not the contribution of this work.

Mechanochromic photonic structures have recently been integrated with camera-based fingertips. Sasso *et al.* demonstrated real-time contact morphology and pressure mapping at approximately 100-μm resolution, including a 5-mm patch in a robotic fingertip [11]. Chen *et al.* subsequently combined a cholesteric liquid-crystal elastomer sensor with a soft gripper for real-time force prediction and closed-loop grasping [12]. Consequently, neither the transduction principle, camera integration, nor closed-loop grasping alone is a defensible contribution. The open robotics question is narrower. We ask whether the spatiotemporal redistribution of structural color provides actionable slip feedback under held-out surface friction, and how its sensing–action trade-off compares with a mechanically and computationally matched GelSight implementation. Predictive trajectory modulation offers a complementary way to avoid slip [13]; here, the arm trajectory is deliberately frozen so that only the gripper feedback channel differs across controllers.

We address this question with shared control on an SO-101 leader–follower platform. The operator supplies or records the arm trajectory, whereas a bounded tactile inner loop modifies only the gripper command. For quantitative comparisons, each controller receives the same frozen arm trajectory, preventing operator motion from becoming a hidden confound. This study makes three contributions:

1. an evaluation design that replays the same `trajectory_source_id` within each paired block and permits tactile feedback to modify only the gripper channel;

2. a dual-fingertip, past-only estimator evaluated by six-fold leave-one-material-out cross-fitting with independent gross-slip labels and frame- and event-level timing metrics; and

3. task-specific robot estimands, intent-to-test sensitivity analysis, and 96 online global-color and delay interventions paired to full photonic-crystal feedback by trajectory, block, repeat, and disturbance.

The words "fast" and "force reduction" are retained only if matched latency measurements and independent command-to-force calibration support them.

2. System and Method

2.1. Shared-Control Architecture

Figure 1 summarizes the question, shared-control boundary, and two robot tasks without assigning measured values to unfinished hardware. The system comprises an SO-101 leader, an SO-101 follower, a two-finger photonic-crystal tactile gripper, a six-axis wrist force/torque sensor, and an external high-speed camera. The leader provides five arm-joint targets and a nominal gripper command. The follower executes the frozen arm trajectory, while the tactile controller may override only the follower gripper channel. The hardware configuration is [HARDWARE: leader/follower servo model and firmware, control rate, calibration IDs, gripper command unit, camera models, and compute platform].

Leader, follower, tactile, wrench, tracking, inference, command-send, and command-ack timestamps are recorded independently. An affine mapping $t_{\text{ref}} = at_{\text{device}} + b$ is estimated for each device and trial block from paired synchronization events. We record offset, drift, residual RMS error, and maximum error. A trial is excluded for synchronization failure only when the maximum mapped residual exceeds one tactile frame.

During pilot work, a leader trajectory is recorded for each object–task–repeat condition. Its identifier and hash are frozen before the main study. The same trajectory is replayed for every controller in the paired block; the operator may start or abort a trial but cannot change its motion. Free teleoperation is used only as a qualitative demonstration. We report [RESULT: leader–follower latency, period distribution, and joint replay error] separately from tactile sensing-to-command latency.

2.2. Photonic-Crystal Fingertips

Each fingertip contains [HARDWARE: layer stack, materials, thicknesses, contact area, optical path, illumination, lens, working distance, field of view, camera frame rate, and effective pixel size]. Two fingertips are mounted on the follower gripper; two additional samples from an independent fabrication batch are reserved for characterization and durability. Cameras warm for 10 min before acquisition, and exposure and white balance remain fixed within a trial block. An unloaded reference image is acquired before each block rather than updated during contact.

For an RGB frame I_t and unloaded reference I_0, we convert both to HSV and encode the hue difference on the unit circle:

$$\mathbf{x}_t = [\sin(2\pi\Delta h_t), \cos(2\pi\Delta h_t) - 1, \Delta s_t, \Delta v_t]. \quad (1)$$

This representation avoids the hue wrap-around discontinuity. Each fingertip is registered to a fixed ROI and resized to 32×32. Left and right frames are paired by nearest timestamp within a locked tolerance and concatenated spatially to give a $4 \times 32 \times 64$ tensor. Unmatched frames are dropped; trials with fewer than the required causal history are invalid rather than imputed. The maximum and 95th-percentile pairing residual are saved for every feature package. [HARDWARE: Low-saturation mask, ROI registration error, illumination normalization, and real dual-camera pairing residuals] are locked before the main dataset.

2.3. Operational Slip Onset and Labels

The high-speed camera tracks rigid markers on the object and both gripper jaws. After subtracting gripper rigid motion and converting pixels to millimeters, gross-slip onset is the first object-to-gripper displacement larger than $\max(0.2\,\text{mm}, 5\sigma)$ for at least 20 ms, where σ is the robust pre-motion tracking noise. A wrench transition provides an independent cross-check; disagreements are adjudicated without viewing tactile predictions. Because local partial-slip ground truth is not measured, we use *slip onset* or *impending gross slip*, not *incipient slip*.

The frame-level impending-slip label uses the 100-ms interval before onset, with a locked 20-ms guard interval. Stable pre-event samples are negative and post-onset frames are excluded. An event trigger is predictive only inside the pre-registered interval $[t_{\text{onset}} - 100\,\text{ms}, t_{\text{onset}})$; earlier triggers are false alarms and post-onset triggers are outside the at-risk set. False alarms per minute use episode start to onset for slip episodes and the full duration for no-slip episodes.

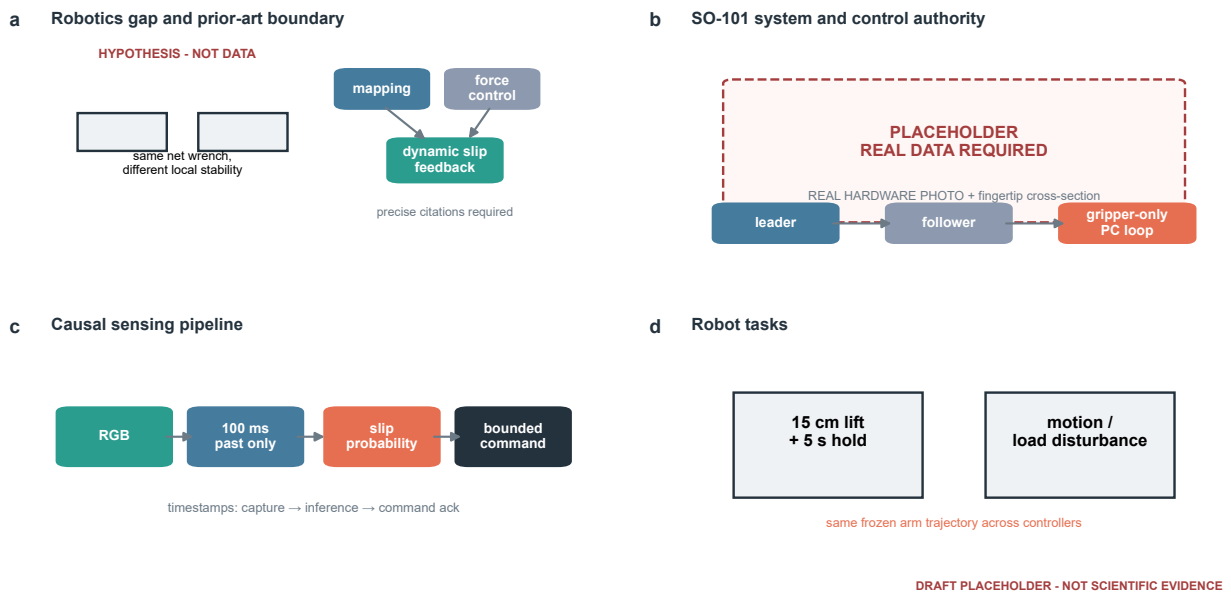


Figure 1. Robotics question and shared-control system. (a) Identical global wrench or gripper commands can conceal different local contact stability; prior work establishes structural-color mapping and force control, while this study tests dynamic slip feedback. (b) Placeholder for the dual-fingertip SO-101 hardware and control authority, where the leader defines arm motion and tactile feedback changes only the gripper channel. (c) Past-only sensing-to-command pipeline. (d) Frozen-trajectory lifting and disturbance tasks. All unmeasured content is explicitly marked as a placeholder and is not scientific evidence.

2.4. Leave-One-Material-Out Cross-Fitting

Six deterministic folds each reserve one material for test and the next material in the locked registry for validation; the remaining four materials are used for training. PC and GelSight have separate models in each fold, but share the fold hash, episode counts, input shape, history, search budget, checkpoint rule, compute platform, and controller. Preprocessing statistics, checkpoint selection, probability calibration, and the threshold targeting a 5% validation false-positive rate are fold-specific and never use the tested material. Each robot object is bound to the model fold that excluded its surface material, and every result records `model_fold_id`, `split_manifest_hash`, and the model package hash.

2.5. Compact Causal Estimator

At time t, the estimator receives only frames timestamped no later than t. For each 100-ms window, it concatenates the final encoded tensor, temporal mean, and final-minus-first change. A fully connected hidden layer with four tanh units produces one slip logit and, when labels are available, contact area and centroid displacement. The matched dual-fingertip configuration contains 98,328 trainable parameters for both PC and GelSight. Auxiliary loss is disabled when contact-geometry labels are absent.

We report frame AUROC, AUPRC, F1, sensitivity, and specificity. Event reporting separates sensitivity, misses, false alarms per at-risk minute, prediction lead for detected events, and inference time. Cross-sensor aggregation verifies fold, split, episode, threshold-source, input-shape, and model provenance before computing paired PC-minus-GelSight effects over the six held-out material folds.

2.6. Bounded Gripper Controller

The controller starts at 10% of the verified normalized command span. Two consecutive probabilities above 0.70 increase the command by 5% of the span. After 0.5 s below 0.30, the command decays at 1% of the span per second and is clamped between 10% and 80%. PC and GelSight share the same control law. A software safety shell holds the last bounded command on non-finite, stale, nonmonotonic, disconnected,

or late inputs. Firmware limits and the physical emergency stop remain independent.

Before an independent calibration, all outcomes are reported as normalized gripper command or servo effort. They are called force in newtons only if [HARDWARE: command-to-force calibration, hysteresis, repeatability, saturation, and calibration ID] are available.

3. Experimental Design

3.1. Sensor Characterization

Four samples from at least two batches are tested at the center and four quadrants. Measurements include the verified nonsaturated range, sensitivity, spatial and sample-to-sample variation, loading–unloading hysteresis, baseline drift, 10–90% step response, and sinusoidal gain and phase from 0.5 to 10 Hz. One sample undergoes 10,000 cycles, followed by the same characterization. Results are [RESULT: sensor characterization table and pre/post durability changes].

3.2. Held-Out-Material Slip Benchmark

The characterization and mechanical-matching contract is shown in Fig. 2. The benchmark crosses six surfaces, three normal-load levels, two tangential speeds, and ten repetitions, yielding 360 PC and 360 GelSight episodes. The matched comparison uses two fingertips per sensor and audits fold hash, training and validation episode counts, 100-ms history, $4 \times 32 \times 64$ input, 98,328 parameters, search budget, checkpoint and threshold rules, compute ID, and controller hash. A fail-closed report prevents aggregation when any field or paired episode identity differs. We report native configurations separately. Contact area, surface friction, elastomer thickness and compliance, mounting geometry, illumination, and native frame rate are [HARDWARE: measured and tabulated]; unmatched properties remain explicit limitations. Offline interventions replace the PC input with global RGB means, a single frame, approximately 1-mm-equivalent sampling, or one-, two-, and four-frame delays. A wrist F/T classifier uses only the pre-registered wrench channels.

3.3. SO-101 Robot Tasks

Figure 3 reserves the exact perception evidence needed to connect the structural-color field to the subsequent robot outcomes. Six 250-g objects span three geometries and six replaceable surfaces. Each controller is evaluated on (i) a 15-cm lift with a 5-s hold and controlled replacement and (ii) recovery from a frozen acceleration/rotation trajectory or a known load step. The four primary conditions are PC temporal feedback, matched GelSight temporal feedback, wrist F/T feedback, and fixed low command. The randomized main study contains $6 \times 4 \times 8 \times 2 = 384$ trials. Eighteen fixed-high reference trials and 96 online intervention trials are analyzed separately. Each global-RGB or two-frame-delay trial reuses the full PC condition's trajectory, block, repeat, initial pose, task, and disturbance identity.

The four primary estimands are: E1, the paired slip-free-completion risk difference between PC and fixed-low control during lifting; E2, the paired restricted-mean time-to-slip difference between PC and wrist F/T during disturbance, with no-slip trials right-censored at task end; E3, their paired maximum-slip difference; and E4, the paired gripper-command time-integral difference between PC and fixed-high control during lifting. Controller order is randomized within paired trajectory blocks. Drops and algorithm failures are outcomes. Only predeclared synchronization, safety-stop, camera, corrupt-file, and precontact failures may be excluded.

3.4. Statistics

The valid synchronized population is primary. The intent-to-test sensitivity assigns incomplete, excluded, aborted, or missing binary outcomes to failure and reports continuous outcomes without imputation together with missingness. Lift and disturbance outcomes are never pooled. Because object and surface are one-to-one, the six tested pairs form six independent clusters. Point estimates are means of within-trajectory paired differences, with object-cluster nonparametric bootstrap 95% intervals. Exact sign-permutation p values are descriptive: with only six clusters, even the smallest four-family Holm-adjusted value cannot reach 0.05. Therefore, inference emphasizes effect sizes, intervals, raw cluster points, and started/valid counts rather than a significance threshold. PC–GelSight comparability is bounded only when the entire 95% interval lies inside the predeclared ± 5 percentage-point success-difference band; nonsignificance is not equivalence.

4. Results

4.1. Sensor Validity and Event-Level Slip Prediction

The verified sensor range, hysteresis, step response, frequency response, spatial variation, and

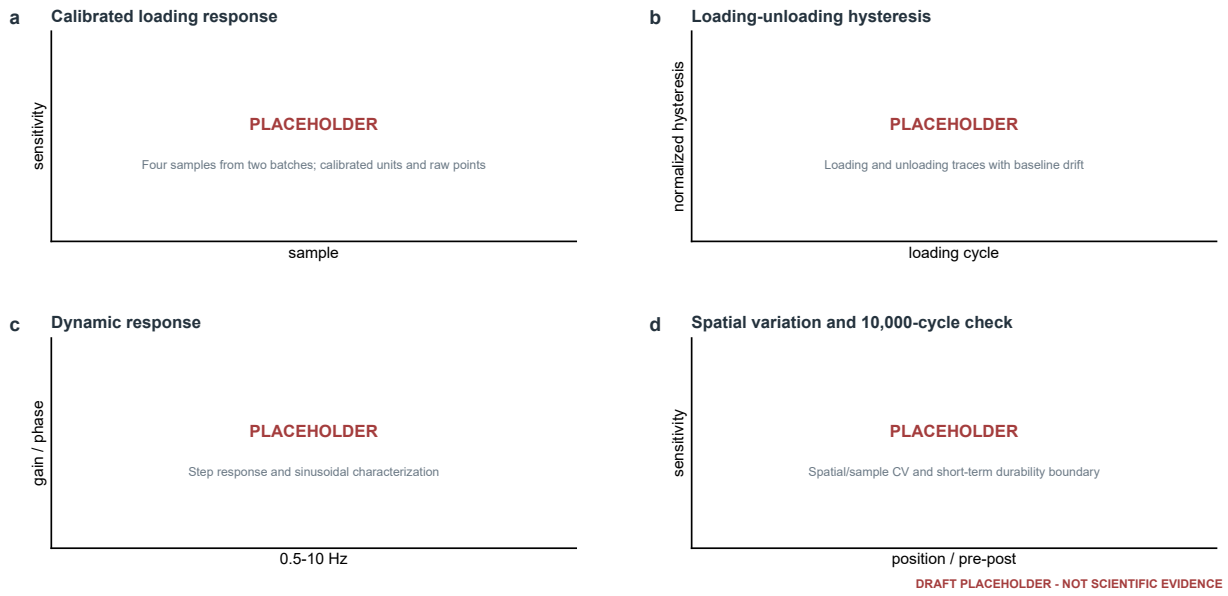


Figure 2. Photonic-crystal fingertip characterization. (a) Sample-level calibrated loading sensitivity for four samples from two batches. (b) Loading–unloading hysteresis. (c) Sinusoidal gain and phase from 0.5 to 10 Hz. (d) The 10,000-cycle pre/post short-term check. Static range, drift, step response, spatial variation, mechanics, synchronization, and operational onset remain in the source tables and system evidence. Placeholder panels contain no measurements.

10,000-cycle pre/post change were [RESULT: characterization estimates, uncertainty, and sample counts]. Across the six cross-fitted held-out surface materials, frame-level performance was [RESULT: AUROC, AUPRC, F1, sensitivity, and specificity by fold]. Event-level reporting yielded [RESULT: sensitivity, misses, false alarms per at-risk minute, and prediction lead with held-out-material-fold intervals]. The real capture rate, dual-fingertip pairing residual, inference time, and exposure-to-command acknowledgement latency were [RESULT: timing measurements].

4.2. Frozen-Trajectory Primary Estimands

Across [RESULT: started, valid synchronized, and intent-to-test counts], E1 estimated [RESULT: PC minus fixed-low slip-free-completion risk difference]. E2 estimated [RESULT: PC minus wrist-F/T restricted-mean time-to-slip difference], including right-censored no-slip completions. E3 and E4 estimated [RESULT: maximum-slip and gripper-command-integral differences]. Each result is reported with raw object–surface cluster points, paired bootstrap intervals, and missingness. No result is pooled across lifting and disturbance tasks.

4.3. Matched PC–GelSight Comparison

The matched audit status was [RESULT: pass/fail with any unmatched fields]. Across the six held-out material folds, paired PC-minus-GelSight perception effects were [RESULT: frame and event effects with fold-level intervals]. Robot-level PC-minus-GelSight effects were [RESULT: task-specific E1–E4-related effects and intervals]. Native configurations are reported separately and do not replace the matched comparison. The success difference is described as bounded comparability only if its entire interval lies within the predeclared interpretation band.

4.4. Trajectory-Paired Interventions and Failure Modes

The 96 online global-RGB and two-frame-delay trials changed [RESULT: task-specific outcomes relative to their paired full-PC trajectories]. Offline single-frame, spatial-resolution, and one-, two-, and four-frame delay interventions changed [RESULT: frame and event metrics]. These interventions test whether specific input information affects the controller outcome; they do not by themselves prove a causal advantage of the photonic-crystal mechanism. Representa-

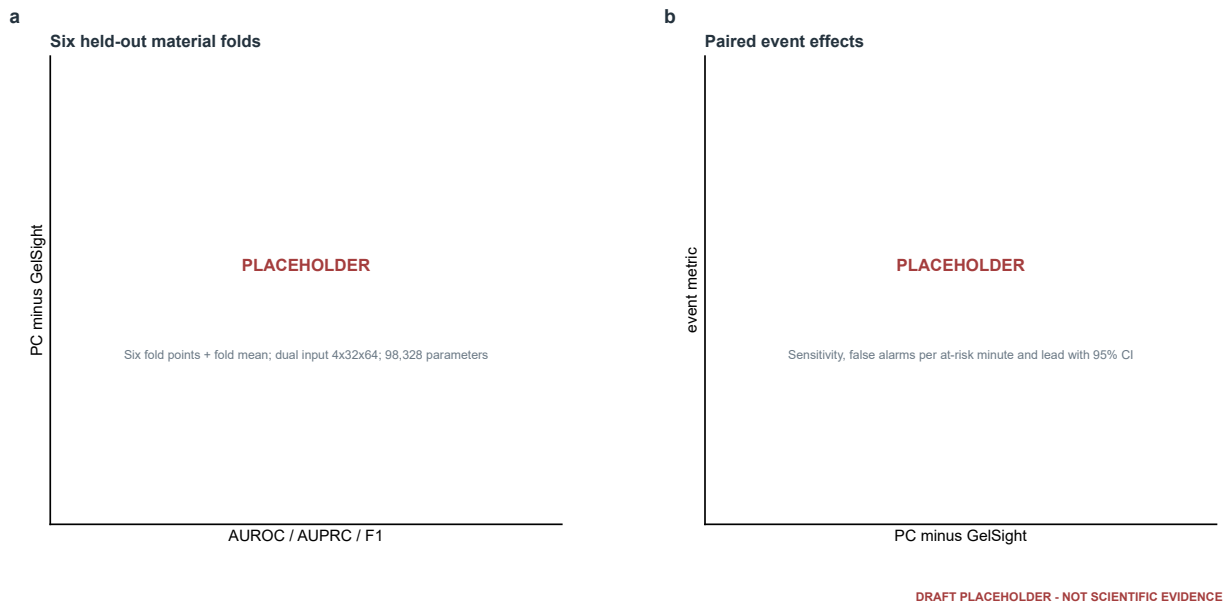


Figure 3. Cross-fitted matched perception comparison. (a) Fold-level PC-minus-GelSight AUROC, AUPRC, and F1 for six held-out surface materials under identical dual-fingertip input shape and model capacity. (b) Paired held-out-material-fold effects for event sensitivity, false alarms per at-risk minute, and prediction lead with 95% intervals. Thresholds are validation-only within each fold. Timing, misses, native settings, and offline interventions are reported in the accompanying table. Placeholder axes contain no synthetic or measured results.

tive failures included [RESULT: genuine failure modes without post-hoc exclusion].

5. Discussion

This study asks whether structural-color dynamics are useful robot feedback, not whether mechanochromic sensing or tactile slip control is new. The decisive evidence is therefore the matched cross-sensor comparison, the frozen-trajectory robot estimands, and the paired intervention pattern. A bounded PC–GelSight interval would support photonic-crystal sensing as a viable alternative in this setting; it would not establish superiority. Spatial and temporal interventions can show that those inputs influence control outcomes, but cannot alone isolate a causal advantage of the photonic-crystal transduction mechanism.

The six-fold design supports held-out evaluation only for the six predefined surface materials. Each robot object is paired with one material, so geometry and surface effects cannot be separated. Six independent object–surface clusters also imply wide intervals and weak p-value resolution; effect sizes and boundaries therefore carry the inference. Frozen replay improves internal validity but does not establish free-teleoperation or human-factor benefits. The LeRobot data and ACT scripts are reproducibility tools, not autonomous-policy evidence.

The evaluation further uses one SO-101 embodiment, fixed internal illumination, flat fingertip geometry, and limited command, speed, material, and disturbance ranges. The external camera defines gross-slip onset and does not resolve local partial slip. A 10,000-cycle test is a short-term durability check rather than lifetime qualification. Servo command or effort is not physical force unless an independent calibration is repeatable over the tested range.

6. Conclusion

We evaluate a photonic-crystal tactile inner loop with independent gross-slip labels, matched sensor baselines, identical replayed arm trajectories, and preregistered information interventions. The system [RESULT: bounded quantitative answer to the event-level perception hypothesis]. In robot trials, it [RESULT: bounded answers for E1–E4 and the intervention hypotheses]. These findings [RESULT: support/do not support] structural-color redistribution as a useful feedback signal within the six tested object–surface pairs;

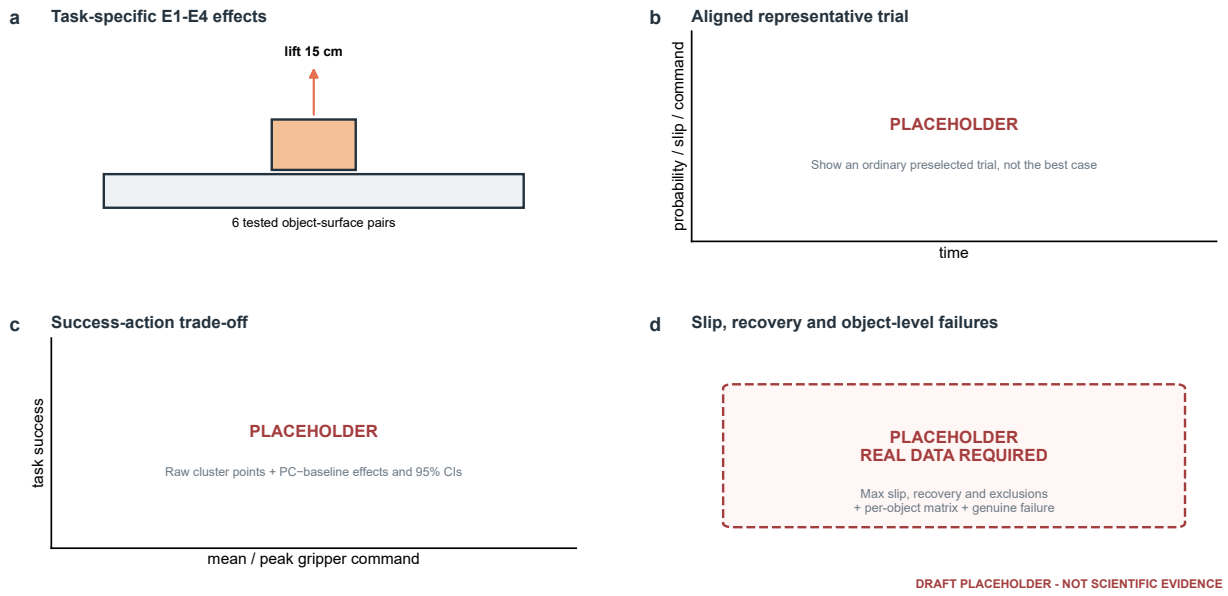


Figure 4. Frozen-trajectory primary estimands and trial evidence. (a) E1–E4 are reported separately for lifting and disturbance. (b) A preselected trace aligns tactile prediction, independent slip onset, and gripper command. (c) Raw points and paired intervals show the six tested object–surface pairs. (d) Started, valid synchronized, intent-to-test, missing, and excluded counts accompany a genuine failure case. No force unit is used without independent calibration; placeholders are not evidence.

they do not establish a new mechanochromic mechanism or generalize beyond the measured platform and contact conditions.

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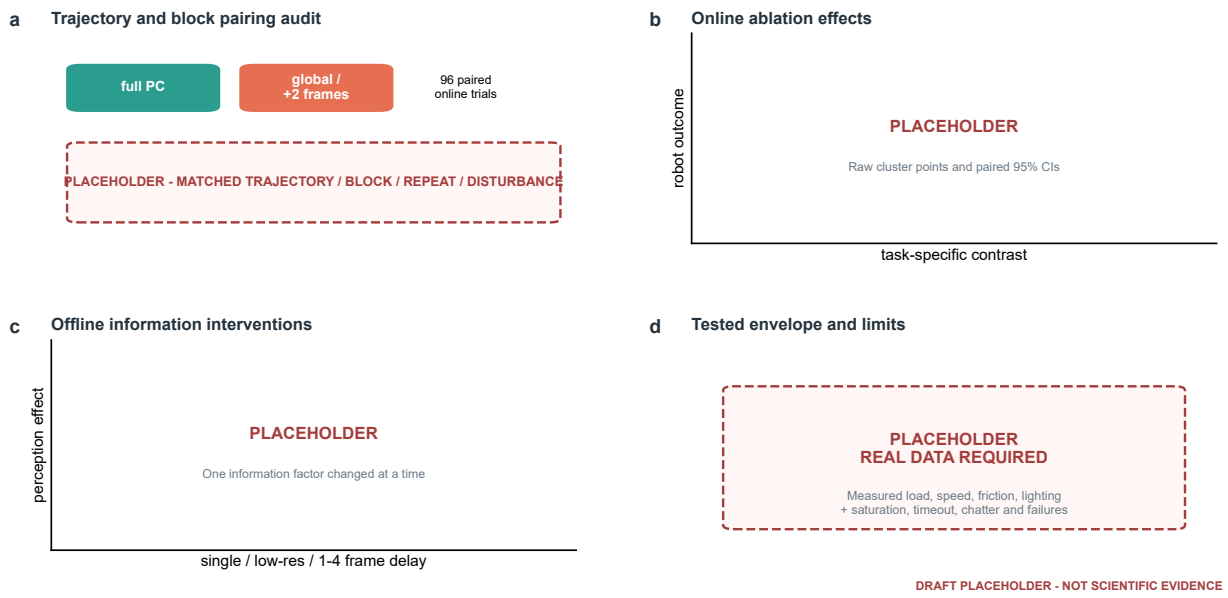


Figure 5. Trajectory-paired information interventions and tested limits. (a) Every online ablation reuses the full-PC trajectory, block, repeat, and disturbance identity. (b) Global-RGB and two-frame-delay robot effects are shown with raw cluster points and paired intervals. (c) Offline temporal, spatial-resolution, and delay effects provide complementary perception evidence. (d) The tested operating envelope and genuine saturation, timeout, chatter, and failure cases delimit the conclusion. Placeholder panels contain no experimental results.

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